NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Which Test Answers the Question Asked?

AP Statistics · Unit 3 · Topic 3.5 · B: 2026-12-03 · E: 2027-01-08

[STAR] TODAY'S PROBLEM

A district of 2,400 students had bus ridership $p_0 = 0.40$ before a new route. Before collecting data, the director asks whether the current proportion p has *increased*. An SRS of 180 current students finds 84 riders.

Maya: "Use $H_0 : p = 0.40$ versus $H_a : p > 0.40$ and a one-sample z-test. Check large counts with 72 and 108."

Theo: "Use $H_a : p \neq 0.40$ because samples can differ either way. Check large counts with the observed 84 and 96."

Current SRS counts

Group	Bus	Other	Total
Students	84	96	180

FIRST TAKE — before the video (5 min)

Does the director want to detect any change, or only more bus riders? Use a phrase from the study.

[BOOK] SET UP THE QUESTION BEFORE USING THE SAMPLE (keep on the page)

Define p as the **population proportion**. For a benchmark p_0, use $H_0 : p = p_0$. Choose $H_a : p > p_0$, $p < p_0$, or $p \neq p_0$ from the research question stated *before* seeing data; $\hat{p}$ is sample evidence and never belongs in the hypotheses. For one categorical variable and one sample, use a one-sample z-test for a population proportion. Verify random data, $n \leq 0.10N$ when sampling without replacement, and the null-expected counts $np_0 \geq 10$ and $n(1 - p_0) \geq 10$.

Finish after the video:

DOK 1 **(a)** Define the population parameter p in context and identify the null value p_0.

DOK 2 **(b)** Compute $\hat{p}$, name the testing procedure, and verify the random, 10 percent, and large-counts conditions using the appropriate values.

DOK 3 **(c)** In 3–4 sentences, choose a setup using the director's word increased. Explain why the count check uses 72 and 108 rather than 84 and 96. State what different question Theo's alternative would answer.

Frame: The warranted alternative is $H_a : p$_____0.40 because the director asked whether ______________________________.

Frame: The large-counts check uses __________ and __________, not ____________, because ________________________________.

EXIT — turn this in

One thing the video changed about my first take: ____